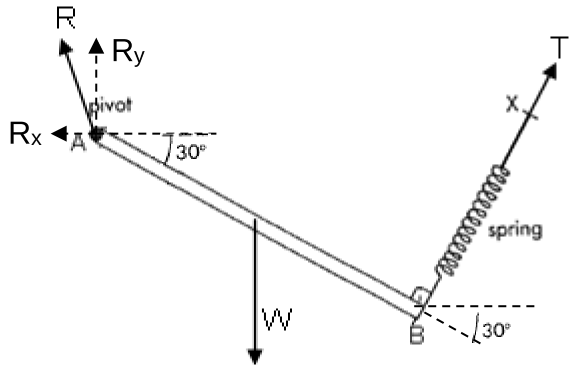
Qns	Answer	Marks
1(a)	Forces must be labelled forces clearly in full – Weight, Reaction Force, Tension Lines of action forces (including intersection of the 3 lines) must be clearly shown.	**1**
(b)	Let *T* be the tension in the spring. Taking moments about A, Sum of clockwise moment = Sum of anticlockwise moment $mg\ (0.20\cos 30°) = T\ (0.40)$ $1.2(9.81)\ (0.20\cos 30°) = T\ (0.40)$ $T = 5.10$ N	**1** **1**

Qns	Answer	Marks
(c)	 Let R_x and R_y be the horizontal and vertical components of the reaction force R. Resolving forces horizontally, $R_x = T\cos 60°$ $= 5.10 \cos 60°$ $= 2.55$ N Resolving forces vertically, $R_y + T\sin 60° = W$ $R_y + 5.10 \sin 60° = (1.2)(9.81)$ $R_y = 7.36$ N magnitude of reaction force $R = \sqrt{R_X^2 + R_Y^2}$ $= \sqrt{2.55^2 + 7.36^2}$ $= 7.79$ N	 1 1 1
(c)	increase (stiffer spring so a smaller extension needed for the same elastic force, resulting in larger angle)	1

Qns	Answer	Marks
2(a)(i)	Total momentum of an isolated system of interacting bodies before	